Software Engineering Practical / Oral Examination

Numerical and Calculation-Based Questions – Solutions

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Questions in this file: Q5, Q9, Q10, Q11, Q18, Q19

Q9, Q10 and Q11 also need a Control Flow Graph – the graph is drawn inside each solution (draw.io files: q9.drawio, q10.drawio, q11.drawio).

All diagram questions are in SE_Practical_Diagram_Questions.docx.

Notes before you start

- Every value is calculated step by step. Intermediate values are kept to 4 decimals; final answers are rounded to 2 decimals.
- KLOC = thousand lines of code (1 KLOC = 1000 LOC). Effort is in **person-months (PM)**; development time is in **months**. They are different things.
- Basic COCOMO: Effort E = a × (KLOC)^b, Time T = c × (E)^d, Average staffing = E / T.
- For Q9–Q11 the pseudo-code in the PDF has lost its indentation; the interpretation used is stated in each solution.

Question 5

1. Question

Q 5: Estimate the development effort and time for a project of size 20 KLOC using the Basic COCOMO model. Assume the project is Organic, with (a=2.4, b=1.05, c=2.5, d=0.38).Calculate the estimated project cost if the development cost is ₹30,000 per person-month.

2. Given Data

- Project size = 20 KLOC
- Project type = Organic (Basic COCOMO)
- a = 2.4, b = 1.05, c = 2.5, d = 0.38
- Cost per person-month = ₹30,000

3. Formula

```
Effort (E) = a × (KLOC)^b   person-months
Development Time (T) = c × (E)^d   months
Average Staffing = E / T   persons
Project Cost = E × Cost per person-month
```

4. Step-by-Step Solution

Step 1: Identify the project size.

```
KLOC = 20
```

Step 2: Calculate (KLOC)^b.

```
(KLOC)^b = 20^1.05
20^1.05 = 23.2317
```

Step 3: Calculate the effort.

```
E = a × (KLOC)^b
E = 2.4 × 23.2317
E = 55.7561 ≈ 55.76 person-months
```

Step 4: Calculate (E)^d.

```
E^d = (55.76)^0.38
E^0.38 = 4.6088
```

Step 5: Calculate the development time.

```
T = c × (E)^d
T = 2.5 × 4.6088
T = 11.5221 ≈ 11.52 months
```

Step 6: Calculate the average staffing.

```
Staffing = E / T
Staffing = 55.76 / 11.52
Staffing = 4.84 ≈ 5 persons
```

Step 7: Calculate the project cost.

```
Project Cost = E × ₹30,000
Project Cost = 55.76 × 30,000
Project Cost = ₹16,72,800
```

5. Final Answers

- **Estimated Effort = 55.76 person-months**
- **Development Time = 11.52 months**
- Average Staffing = 4.84 persons (about 5)
- **Project Cost = ₹16,72,800** (≈ ₹16.73 lakh)

6. Interpretation

About 5 people working for about 11.5 months (≈ 1 year) are needed to build a 20 KLOC organic project, costing roughly ₹16.7 lakh.

7. Verification

- Formulas checked: Basic COCOMO organic mode with the given constants.
- Units: KLOC used directly (not LOC); effort in PM; time in months.
- Cost using the unrounded effort (55.7561 PM) is ₹16,72,684; the small difference from ₹16,72,800 is only due to rounding E to 2 decimals. Both ≈ ₹16.73 lakh.

Question 9

1. Question

Q 9. 1. Apply Basis Path Testing to the given program and calculate the Cyclomatic Complexity.

```
read n

if (n > 0)

   if (n % 2 == 0)

       print "Even"

   else

       print "Odd"

else

   print "Invalid"
```

2. Given Data

- Program: read n; if (n > 0) { if (n % 2 == 0) print "Even" else print "Odd" } else print "Invalid"
- Interpretation: the inner if–else belongs to the true branch of the outer if (n > 0); the last else belongs to the outer if.
- Method: Basis Path Testing (white-box testing)

3. Formula

```
Cyclomatic Complexity  V(G) = E − N + 2   (E = edges, N = nodes)
V(G) = P + 1   (P = number of predicate / decision nodes)
V(G) = number of regions of the CFG (including the outside region)
```

The value of V(G) = number of independent paths in a basis set = minimum number of test cases for full branch coverage.

4. Step-by-Step Solution

Step 1: Number the statements (nodes of the CFG).

Node	Statement
1	read n
2	if (n > 0)
3	if (n % 2 == 0)
4	print "Even"
5	print "Odd"
6	print "Invalid"
7	End (exit node)

Step 2: Draw the Control Flow Graph (CFG).

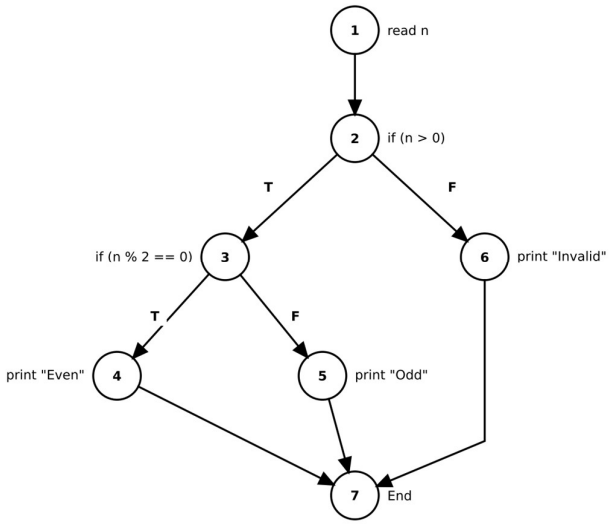


Figure: Control Flow Graph (T = true branch, F = false branch)

Step 3: List the edges.

```
1 → 2
2 → 3 (T)
2 → 6 (F)
3 → 4 (T)
3 → 5 (F)
4 → 7
5 → 7
6 → 7
```

Step 4: Count nodes, edges and predicate nodes.

```
Nodes N = 7
Edges E = 8
Predicate nodes P = 2
```

Step 5: Calculate V(G) by all three methods.

```
Method 1: V(G) = E − N + 2 = 8 − 7 + 2 = 3
Method 2: V(G) = P + 1 = 2 + 1 = 3
Method 3: Regions = 3 (the enclosed areas of the graph + the outer area)
```

Step 6: Write the independent paths (basis path set).

Path	Nodes
P1	1 – 2 – 3 – 4 – 7
P2	1 – 2 – 3 – 5 – 7
P3	1 – 2 – 6 – 7

Step 7: Design test cases (one per path).

Test case	Input	Path covered	Expected output
TC1	n = 4	P1	Even
TC2	n = 7	P2	Odd
TC3	n = −5 (or 0)	P3	Invalid

5. Final Answers

- **Cyclomatic Complexity V(G) = 3**
- Number of independent paths = 3: P1 (Even), P2 (Odd), P3 (Invalid)
- Minimum test cases needed = 3

6. Interpretation

V(G) = 3 means the program is simple (V(G) ≤ 10, low risk) and 3 test cases are enough to execute every branch once.

7. Verification

- E − N + 2 = 8 − 7 + 2 = 3; P + 1 = 2 + 1 = 3 – both methods agree.
- Every edge appears in at least one path; each new path introduces one new edge not used earlier (3→4, 3→5, 2→6).

Question 10

1. Question

Q 10: Apply White-box testing and calculate cyclomatic complexity for given program

```
read n

i = 2; isPrime = true

while (i <= n/2)

if (n % i == 0)

isPrime = false

break

i = i + 1;print isPrime
```

2. Given Data

- Program: checks whether n is prime.
- **Issue in the question:** the indentation is missing and 'i = i + 1;print isPrime' is on one line. Assumption: the while-body = { if (n % i == 0) { isPrime = false; break } ; i = i + 1 } and 'print isPrime' is executed once after the loop.
- n/2 is integer division (e.g. 5/2 = 2).
- Method: White-box (basis path) testing

3. Formula

```
Cyclomatic Complexity  V(G) = E – N + 2   (E = edges, N = nodes)
V(G) = P + 1   (P = number of predicate / decision nodes)
V(G) = number of regions of the CFG (including the outside region)
```

The value of V(G) = number of independent paths in a basis set = minimum number of test cases for full branch coverage.

4. Step-by-Step Solution

Step 1: Number the statements (nodes of the CFG).

Node	Statement
1	read n; i = 2; isPrime = true
2	while (i <= n/2)
3	if (n % i == 0)
4	isPrime = false; break
5	i = i + 1
6	print isPrime (exit)

Step 2: Draw the Control Flow Graph (CFG).

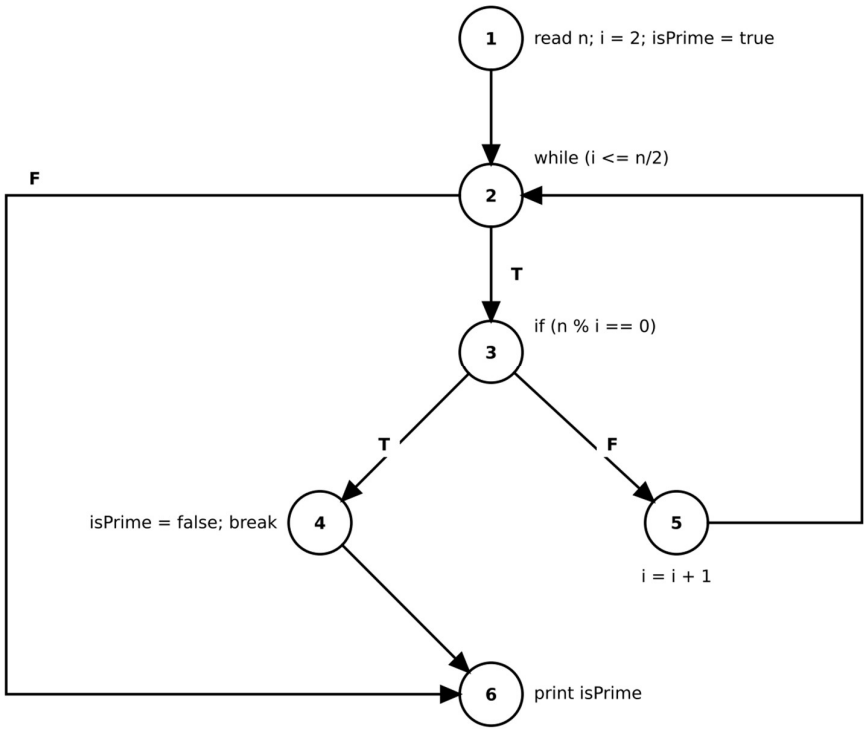


Figure: Control Flow Graph (T = true branch, F = false branch)

Step 3: List the edges.

```
1 → 2
2 → 3 (T: loop condition true)
2 → 6 (F: loop ends)
3 → 4 (T: divisor found)
3 → 5 (F)
4 → 6 (break leaves the loop)
5 → 2 (back edge to the loop condition)
```

Step 4: Count nodes, edges and predicate nodes.

```
Nodes N = 6
Edges E = 7
Predicate nodes P = 2
```

Step 5: Calculate V(G) by all three methods.

```
Method 1: V(G) = E – N + 2 = 7 – 6 + 2 = 3
Method 2: V(G) = P + 1 = 2 + 1 = 3
Method 3: Regions = 3 (the enclosed areas of the graph + the outer area)
```

Step 6: Write the independent paths (basis path set).

Path	Nodes
P1	1 – 2 – 6 (loop not entered)
P2	1 – 2 – 3 – 4 – 6 (divisor found, break)
P3	1 – 2 – 3 – 5 – 2 – 6 (one iteration, no divisor, loop ends)

Step 7: Design test cases (one per path).

Test case	Input	Path covered	Expected output
TC1	n = 3 (n/2 = 1, so 2 ≤ 1 is false)	P1	isPrime = true
TC2	n = 4 (4 % 2 = 0)	P2	isPrime = false
TC3	n = 5 (n/2 = 2; 5 % 2 ≠ 0; i = 3, 3 ≤ 2 false)	P3	isPrime = true

5. Final Answers

- **Cyclomatic Complexity V(G) = 3**
- Independent paths = 3 (P1, P2, P3); minimum test cases = 3

6. Interpretation

V(G) = 3: the program is simple. The two decisions (while and if) each add one to the complexity. The loop adds a back edge, but it is still counted by the formula.

7. Verification

- E − N + 2 = 7 − 6 + 2 = 3; P + 1 = 2 + 1 = 3 – both agree.
- If the assumption about the loop body changes (for example i = i + 1 outside the loop), the graph changes – so the assumption is stated above.

Question 11

1. Question

Q 11: Given the following pseudo-code, apply Basis Path Testing to determine:

```
read a, b

if (a > b)

max = a

else

max = b

if (max > 0)

print "Positive"

else

print "Non-positive"
```

1. Control Flow Graph (CFG)
2. Cyclomatic Complexity
3. Independent Paths
4. Test Cases

2. Given Data

- Program: finds the larger of a and b, then tells whether it is positive.
- Two sequential if–else statements.
- Method: Basis Path Testing

3. Formula

```
Cyclomatic Complexity  V(G) = E – N + 2   (E = edges, N = nodes)
V(G) = P + 1   (P = number of predicate / decision nodes)
V(G) = number of regions of the CFG (including the outside region)
```

The value of V(G) = number of independent paths in a basis set = minimum number of test cases for full branch coverage.

4. Step-by-Step Solution

Step 1: Number the statements (nodes of the CFG).

Node	Statement
1	read a, b
2	if (a > b)
3	max = a
4	max = b
5	if (max > 0)
6	print "Positive"
7	print "Non-positive"
8	End (exit node)

Step 2: Draw the Control Flow Graph (CFG).

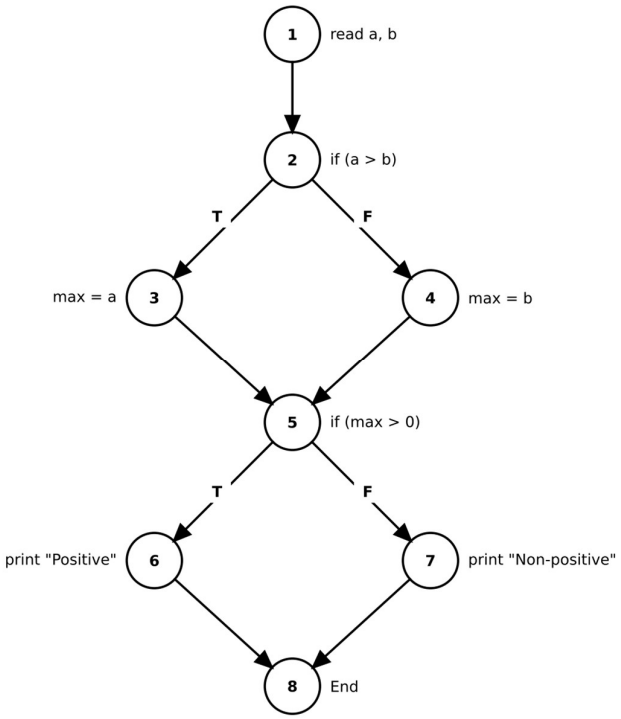


Figure: Control Flow Graph (T = true branch, F = false branch)

Step 3: List the edges.

```
1 → 2
2 → 3 (T)
2 → 4 (F)
3 → 5
4 → 5
5 → 6 (T)
5 → 7 (F)
6 → 8
7 → 8
```

Step 4: Count nodes, edges and predicate nodes.

```
Nodes N = 8
Edges E = 9
Predicate nodes P = 2
```

Step 5: Calculate V(G) by all three methods.

```
Method 1: V(G) = E – N + 2 = 9 – 8 + 2 = 3
Method 2: V(G) = P + 1 = 2 + 1 = 3
Method 3: Regions = 3 (the enclosed areas of the graph + the outer area)
```

Step 6: Write the independent paths (basis path set).

Path	Nodes
P1	1 – 2 – 3 – 5 – 6 – 8 (baseline path)
P2	1 – 2 – 4 – 5 – 6 – 8 (changes the first decision)
P3	1 – 2 – 3 – 5 – 7 – 8 (changes the second decision)

Step 7: Design test cases (one per path).

Test case	Input	Path covered	Expected output
TC1	a = 5, b = 3	P1	max = 5, Positive
TC2	a = 2, b = 4	P2	max = 4, Positive
TC3	a = −2, b = −5	P3	max = −2, Non-positive

Note: the fourth possible path 1 – 2 – 4 – 5 – 7 – 8 (for example a = −5, b = −2 gives Non-positive) is NOT independent – it is a combination of the basis paths: P2 + P3 − P1. Since V(G) = 3, a basis set needs only 3 paths.

5. Final Answers

- 1. CFG: drawn above (8 nodes, 9 edges)
- 2. **Cyclomatic Complexity V(G) = 3**
- 3. Independent paths: P1 = 1-2-3-5-6-8, P2 = 1-2-4-5-6-8, P3 = 1-2-3-5-7-8
- 4. Test cases: TC1 (5, 3), TC2 (2, 4), TC3 (−2, −5)

6. Interpretation

V(G) = 3 means low complexity. Although there are 4 possible routes through the code (2 × 2), only 3 are linearly independent; testing the 3 basis paths covers every edge.

7. Verification

- E − N + 2 = 9 − 8 + 2 = 3; P + 1 = 2 + 1 = 3 – both agree.
- All 9 edges are covered by P1, P2 and P3 together: P1 (1-2, 2-3, 3-5, 5-6, 6-8), P2 adds 2-4 and 4-5, P3 adds 5-7 and 7-8.

Question 18

1. Question

Q. 18. A healthcare startup plans to develop an Online Pharmacy Management System to enable customers to search medicines, upload prescriptions, place orders, make online payments, and track deliveries. The system will also allow pharmacists to verify prescriptions, manage inventory, and generate sales reports.

The estimated software size is 32 KLOC. Based on previous projects, the organization classifies it as a Semi-Detached project.

The function counts identified are:

- External Inputs (EI): 18
- External Outputs (EO): 14
- External Inquiries (EQ): 10
- Internal Logical Files (ILF): 8
- External Interface Files (EIF): 5

Assume all functions are of average complexity and the Value Adjustment Factor (VAF) = 1.05.

Questions:

1. Calculate the Unadjusted Function Points (UFP).
2. Calculate the Adjusted Function Points (AFP).
3. Calculate the project effort using the Basic COCOMO model. a=2.4, b=1.05, c=2.5, d=0.38
4. Find the estimated effort and discuss whether the project is suitable for the selected COCOMO category.

2. Given Data

- Estimated size = 32 KLOC
- Project class (as given) = Semi-Detached
- EI = 18, EO = 14, EQ = 10, ILF = 8, EIF = 5 (all average complexity)
- VAF = 1.05
- a = 2.4, b = 1.05, c = 2.5, d = 0.38 (constants given in sub-question 3)

Inconsistency in the question (please note): the constants a = 2.4, b = 1.05, c = 2.5, d = 0.38 are the standard constants of the **Organic** mode, but the project is called **Semi-Detached**. The standard Semi-Detached constants are a = 3.0, b = 1.12, c = 2.5, d = 0.35. As the question explicitly gives the constants, they are used for the main answer; the standard Semi-Detached values are shown in Part 6 for comparison. The 32 KLOC size is given directly, so the function points are not converted to KLOC.

3. Formula

```
Average-complexity weights: EI = 4, EO = 5, EQ = 4, ILF = 10, EIF = 7
UFP = Σ (count × weight)
AFP = UFP × VAF
E = a × (KLOC)^b ;  T = c × (E)^d ;  Staffing = E / T
```

4. Step-by-Step Solution

Part 1: Unadjusted Function Points (UFP)

Function type	Count	Weight (average)	Count × Weight

EI	18	4	18 × 4 = 72
EO	14	5	14 × 5 = 70
EQ	10	4	10 × 4 = 40
ILF	8	10	8 × 10 = 80
EIF	5	7	5 × 7 = 35

```
UFP = 72 + 70 + 40 + 80 + 35
UFP = 297
```

Part 2: Adjusted Function Points (AFP)

```
AFP = UFP × VAF
AFP = 297 × 1.05
AFP = 311.85
```

Part 3: Basic COCOMO effort (with the given constants)

```
Step 1: KLOC = 32
Step 2: 32^1.05 = 38.0546
Step 3: E = a × (KLOC)^b
        E = 2.4 × 38.0546
        E = 91.3311 ≈ 91.33 person-months
Step 4: E^0.38 = 5.5595
Step 5: T = c × (E)^d
        T = 2.5 × 5.5595
        T = 13.8987 ≈ 13.90 months
Step 6: Average staffing = 91.33 / 13.90 = 6.57 ≈ 7 persons
```

Part 4: Suitability for Semi-Detached category (comparison)

```
Standard Semi-Detached constants: a = 3.0, b = 1.12, c = 2.5, d = 0.35
32^1.12 = 48.5029
E = 3.0 × 48.5029 = 145.51 person-months
E^0.35 = 5.7149
T = 2.5 × 5.7149 = 14.29 months
Average staffing = 145.51 / 14.29 = 10.18 persons
```

5. Final Answers

- **UFP = 297 function points**
- **AFP = 311.85 function points**
- **Estimated Effort (given constants) = 91.33 person-months**
- Development Time = 13.90 months; average staffing ≈ 6.6 persons
- For comparison, standard Semi-Detached constants give Effort = 145.51 person-months, Time = 14.29 months

6. Interpretation

The system has several different parts (prescription verification, payments, delivery tracking, inventory) and the team has mixed experience, which fits the **Semi-Detached** category (medium size, mixed team and requirements). However, the constants given in the question are Organic-mode constants, so the result of 91.33 person-months is probably an **under-estimate**. With the proper Semi-Detached constants the effort is about 146 person-months (1.59 times higher). So: the project is suitable for the Semi-Detached category, but the given constants are not; use a = 3.0, b = 1.12 for a realistic estimate.

7. Verification

- UFP check: 72 + 70 + 40 + 80 + 35 = 297.
- AFP check: 297 × 1.05 = 311.85.
- KLOC (32) used, not LOC; effort is in person-months and time in months.
- E/T = 91.33/13.90 = 6.57.
- The mismatch between the 'Semi-Detached' class and the Organic constants is flagged above and not silently changed.

Question 19

1. Question

Q. 19. A municipal corporation plans to develop a Smart Parking Management System that allows users to locate available parking spaces, reserve slots, make online payments, and receive real-time parking updates. Parking administrators can monitor occupancy and generate daily reports.

The estimated project size is 28 KLOC, and the project is categorized as Organic. a=2.4, b=1.05, c=2.5, d=0.38

Questions:

1. Calculate the project effort and development time using the Basic COCOMO model.
2. If the organization's productivity is 600 LOC per person-month, Calculate the effort using the LOC method.

2. Given Data

- Project size = 28 KLOC = 28,000 LOC
- Project type = Organic
- a = 2.4, b = 1.05, c = 2.5, d = 0.38
- Productivity = 600 LOC per person-month

3. Formula

```
Basic COCOMO:  E = a × (KLOC)^b ;  T = c × (E)^d ;  Staffing = E / T
LOC method:  Effort = Size (LOC) / Productivity (LOC per person-month)
```

4. Step-by-Step Solution

Part 1: Basic COCOMO

```
Step 1: KLOC = 28
Step 2: 28^1.05 = 33.0762
Step 3: E = a × (KLOC)^b
        E = 2.4 × 33.0762
        E = 79.3829 ≈ 79.38 person-months
Step 4: E^0.38 = 5.2710
Step 5: T = c × (E)^d
        T = 2.5 × 5.2710
        T = 13.1776 ≈ 13.18 months
Step 6: Average staffing = 79.38 / 13.18 = 6.02 ≈ 6 persons
```

Part 2: LOC method

```
Step 1: Convert KLOC to LOC:  28 KLOC = 28 × 1000 = 28,000 LOC
Step 2: Effort = Size / Productivity
Step 3: Effort = 28,000 / 600
Step 4: Effort = 46.67 person-months
```

5. Final Answers

- **Effort (Basic COCOMO) = 79.38 person-months**
- **Development Time = 13.18 months**
- Average staffing = 6.02 persons (about 6)
- **Effort (LOC method) = 46.67 person-months**

6. Interpretation

The LOC method gives a smaller effort (46.67 PM) than COCOMO (79.38 PM). The reason: COCOMO implies a productivity of 28,000 / 79.38 = 353 LOC per person-month, while the organisation's stated productivity is higher (600 LOC/PM). If the organisation really achieves 600 LOC/PM, the LOC estimate is more specific to it; COCOMO is a general industry model.

7. Verification

- 28 KLOC converted to 28,000 LOC for the LOC method; KLOC (28) used in COCOMO.
- E/T = 79.38/13.18 = 6.02.
- 28,000 / 600 = 46.67 (600 × 46.67 = 28,002 ≈ 28,000).
- The question does not ask for development time or cost for the LOC method, so none is given.